Journal of Applied Physics

LETTER | MAY 15 1987

O_2 chemical potential of nonstoichiometric ceria, CeO_{2-x}, determined by a solid electrochemical method

I. Riess; H. Janczikowski; J. Nölting

Check for updates

J. Appl. Phys. 61, 4931–4933 (1987)
https://doi.org/10.1063/1.338363

View Online
Export Citation

Articles You May Be Interested In

Coulometric trace humidity measurement in technical gases

Rev. Sci. Instrum. (August 2018)

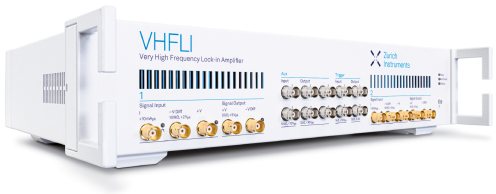
Development of microcoulometry for measuring oxygen content in copper oxides

Appl. Phys. Lett. (December 2005)

Accurate oxygen-content determination method for decreased sample amounts of superconductive and other functional oxides

Appl. Phys. Lett. (July 2002)

19 March 2026 09:52:58



Zurich Instruments

Freedom to Innovate.

The New VHFLI 200 MHz Lock-in Amplifier.

Orchestrate pulses, triggers, and acquisition as the hub of your experiment.
Discover more – run every signal analysis tool, simultaneously.

Order now

AIP Publishing

O_2 chemical potential of nonstoichiometric ceria, CeO_{2-x}, determined by a solid electrochemical method

I. Riess
Department of Physics, Technion-Israel Institute of Technology, Haifa 32000, Israel

H. Janczikowski and J. Nölting
Institute for Physical Chemistry, Göttingen University and Sonderforschungsbereich 126 Göttingen-Clausthal, Tammann Strasse 6, D-3400 Göttingen, Federal Republic of Germany

(Received 21 January 1986; accepted for publication 13 January 1987)

The chemical potential of oxygen in nonstoichiometric ceria, CeO_{2-x}, was measured versus composition x ($10^{-4} \leqslant x \leqslant 10^{-2}$) and temperature ($823 \leqslant T \leqslant 1073$ K) using a novel solid electrochemical cell, utilizing yttrium stabilized zirconia. The composition of CeO_{2-x} was varied by coulometric titration. The O_2 chemical potential was determined from the electromotive force of the cell with air used as the reference gas. With this new experimental setup, it is possible to carry out long coulometric titrations at very low oxygen partial pressures (limited by 10^{-30} atm at 823 K or by 10^{-13} atm at 1473 K). For x between 10^{-4} and 10^{-2}, the O_2 partial pressure in equilibrium with CeO_{2-x} was found to change from 10^{-8} to 10^{-18} atm at 1073 K and from 10^{-16} to 10^{-30} atm at 823 K. The oxygen vacancies in CeO_{2-x} were determined to be V_O'' in this (x,T) range. The charge on aliovalent cations impurities was found to dominate the negative (effective) charge concentration for $x \lesssim 10^{-3}$. The oxygen partial molar enthalpy $\Delta\bar{H}(O_2)$ is 870 kJ/mol (9 eV) for $x \gtrsim 3 \times 10^{-3}$ decreasing for smaller x to a value of 670 kJ/mol (7 eV) for $x = 10^{-4}$.

In investigating thermodynamic properties of an oxide, it is essential to determine the chemical potential $\mu(O_2)$ of oxygen, and to control and measure the oxygen concentration in the solid. An accurate and convenient method to determine these two quantities is by means of solid-state electrochemical methods.

To accomplish the electrochemical measurement, a solid oxygen conductor (SE) conducting O^{2-} ions is placed between the oxide and a reference gas, containing oxygen with a given partial pressure, $P(O_2)$.[1] The chemical potential $\mu(O_2)$ is obtained from the voltage signal (EMF) which appears on the SE,

$$\mathrm{EMF} = \mu(O_2) - \mu^{\mathrm{ref}}(O_2)/4F, \qquad (1)$$

where F is the Faraday constant and $\mu(O_2)^{\mathrm{ref}}$ is the chemical potential of oxygen in the reference gas. The temperature T and the composition of the oxide can be fixed *in situ*. The latter is done very accurately by passing a known charge through the electrochemical cell (coulometric titration). An adequate solid electrolyte for such measurements is Y_2O_3-doped ZrO_2 (YSZ) which conducts O^{2-} ions. To assure a dominant ionic conductivity, YSZ should be operated well within its electrolytic domain[2] [e.g., $1 \geqslant P(O_2) \gtrsim 10^{-30}$ atm at 829 K and $1 \geqslant P(O_2) \gtrsim 10^{-13}$ atm at 1473 K[3]], where for practical reasons $P(O_2)$ above 1 atm is not considered.

In the past, attempts to build a solid electrochemical cell based on YSZ revealed that the major problem was O_2 leakage through unavoidable fine cracks in the SE ceramic YSZ.[4] The leakage introduces uncontrolled and excessive changes in the oxide composition. The arrangement proposed by Tretyakov and Rapp[4] could be operated with limited leakage with $P(O_2)$ differences of less than four orders of magnitude. It was later suggested to reduce the leakage by maintaining $P(O_2)$ at the surrounding of the sample compartment as close as possible to that inside the compartment.[5–7]

We have constructed a novel cell for coulometric titration and EMF measurements on oxides enabling the determination of $\mu(O_2)$ from $P(O_2) = 10^{-8}$ atm down to $P(O_2) = 10^{-30}$ atm while staying within the aforementioned electrolytic domain. A wide range of reference $P(O_2)$ values, including room air, can be used.

The cell consists of two YSZ tubes, one for the sample (a) and the other for the reference gas (b), in Fig. 1. A YSZ pellet (P) placed between the two tube permits ionic conduction between the two compartments. The pellet is needed to eliminate direct diffusion of O_2 molecules from (b) to (a) through adjacent fine cracks in the YSZ tubes. A stream of purified Ar [$P(O_2) = 10^{-16}$–10^{-20} atm, $P(H_2) < 10^{-5}$ atm] is passed inside an Al_2O_3 tube (c) from the sample side to the reference side, at 1 cm^3/s, to flush the YSZ tubes. The inside of the sample compartment (a) is held at a slight

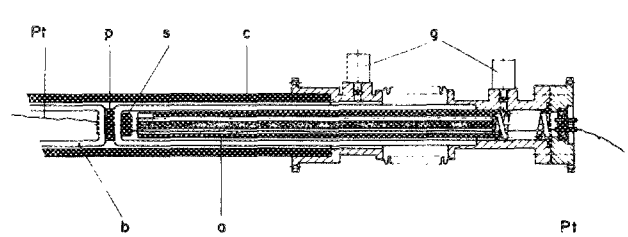


FIG. 1. Electrochemical cell with two separated compartments. a, b: separated compartments; c: alumina tube; g: gas inlets; p: yttria-doped zirconia pellet; s: CeO_{2-x} sample; Pt platinum leads which end at porous Pt electrodes.

19 March 2026 09:52:58

 0021-8979/87/104931-03$02.40 © 1987 American Institute of Physics

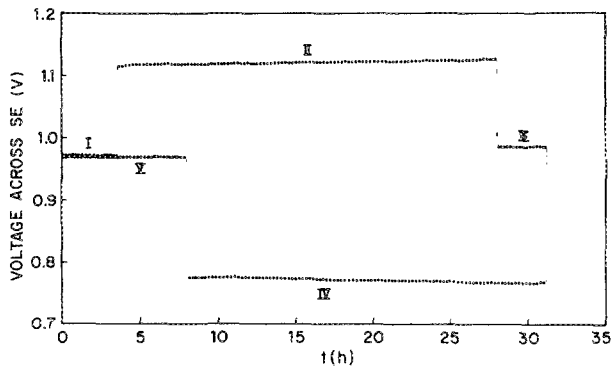


FIG. 2. Cyclic coulometric titration for leakage test of the electrochemical setup.

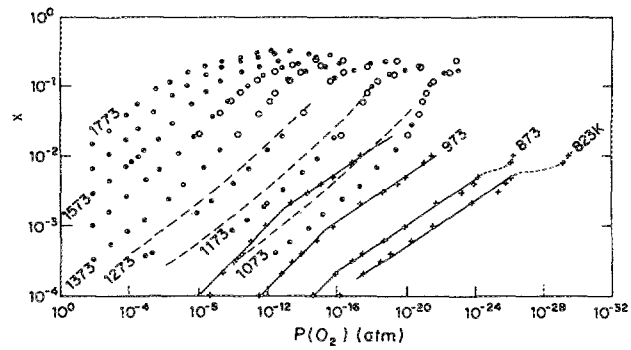


FIG. 4. Oxygen equilibrium pressure of CeO_{2-x} vs oxygen deficiency x. Dashed line: Tuller and Nowick. ●: Panlener *et al.* O: Bevan and Kordis. + : this work (experimental). Solid line: this work (theoretical).

overpressure (1.01 atm) with purified and stationary Ar. The reference compartment (b) is continuously flushed with air. The electrodes on the SE and the sample(s) are porous Pt layers. The cell is placed in an oven and positioned to eliminate a temperature difference between the electrodes.

Sample composition was limited to keep $P(O_2)$ within the electrolyte domain of YSZ and also below 10^{-8} atm to minimize O_2 leakage from the sample compartment into the Ar stream. Measurements of $\mu(O_2)$ as a function of composition and temperature were carried out on nonstoichiometric cerium oxide, CeO_{2-x}, in the high-temperature α phase of CeO_{2-x} (phase boundaries were taken from Ref. 8).

The absolute value of x was determined as follows: The sample was first brought to the state with $P(O_2) = 2\times 10^{-9}$ atm and $T = 1073$ K and x was determined by extrapolation from the data in Ref. 9. Then x was changed by coulometric titration and the changes in x due to the titration were added to the initial value of x.

To demonstrate, first, the accurate and leak-free performance of the new experimental setup, the following experiment at 1073 K was carried out (see Fig. 2):

(I) The composition of the CeO_{2-x} sample was brought to $x \sim 10^{-2}$ by coulometric titration. The EMF of 971 mV is observed to be stable within less than 1 mV over a period of ~24 h (only part of which is shown in Fig. 2), indicating a leak-free system and a constant composition of the probe.

(II) A second coulometric titration was carried out over a period of ~24 h.

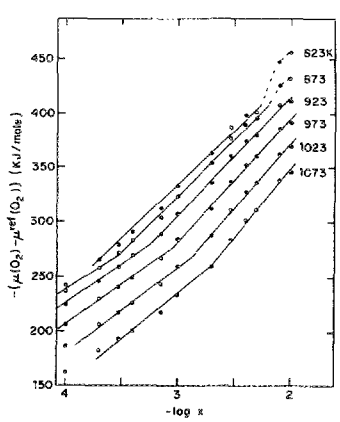


FIG. 3. Chemical potential of oxygen vs oxygen deficiency x.

(III) After the coulometric titration was terminated the probe reached equilibrium within 15 min and remained at a constant EMF (13 mV higher than at stage I).

(IV) A third reverse coulometric titration with equal charge as in period II is applied.

(V) A stable EMF is observed which deviates less than 2 mV from that of the beginning. This corresponds to an error in composition $x = 10^{-4}$ and $\Delta x/x = 10^{-2}$ after the 62-h lasting cyclic experiment.

$\mu(O_2)$, the chemical potential of oxygen in equilibrium with the sample CeO_{2-x} at a partial pressure $P(O_2)$ is

$$\mu(O_2) = \mu^{\phi}(O_2) + RT\ln P(O_2)$$
$$= \mu^{\mathrm{ref}}(O_2) + RT\ln P(O_2)/P^{\mathrm{ref}}(O_2)\,, \qquad (2)$$

where R is the gas constant. $\mu(O_2) - \mu^{\mathrm{ref}}(O_2)$ is obtained from the EMF according to Eq. (1). The composition is determined by the coulometric titration. Figure 3 experimentally presents determined values of $\mu(O_2) - \mu^{\mathrm{ref}}(O_2)$ versus the deficiency in oxygen x for the temperature range 823–1073 K. The theory of low defect concentration in nonstoichiometric CeO_{2-x} predicts[9]

$$\mu(O_2) - \mu^{\mathrm{ref}}(O_2) = -6RT\ln x + \mathrm{const.} \qquad (3)$$

Equation (3) holds for an undoped CeO_{2-x} and for dominant oxygen vacancies (V_O'') doubly charged with respect to the lattice. The solid straight lines for the higher x value were drawn in Fig. 3 according to Eq. (3).

For lower x the impurities in a concentration of 900 ppm[10] cannot be neglected. If aliovalent cations in CeO_{2-x} are the dominant negative (effective) charges and the oxygen vacancies are V_O'' then the theory predicts[9]

$$\mu(O_2) - \mu(O_2) = -4RT\ln x + \mathrm{const.} \qquad (4)$$

The solid straight lines for the lower x values were drawn in Fig. 3 according to Eq. (4).

Combining Eqs. (2) and (3) or (4) yields linear relations between $\ln P(O_2)$ and $\ln x$. We have added our results for the $P(O_2)$ vs x relation, to those obtained by others[9,11,12] and presented them in Fig. 4. There is a quantitative agreement of our data for $T = 1073$ K with that reported by Tuller and Nowick,[9] where x was determined (indirectly) from conductivity and mobility measurements. The present x values are, however, up to a factor of 3.5 higher than those determined by thermogravimetry.[11–13] The slope (for

19 March 2026 09:52:58

$T = 1073$ K) is also not in agreement, as seen in Fig. 4. We find $\ln x = -(1/4)\ln P(O_2) + \text{const}$ for $10^{-4} \lesssim x \lesssim 10^{-3}$ and $\ln k = -(1/6)\ln P(O_2) + \text{const}$ for $10^{-3} \lesssim x \lesssim 10^{-2}$.

The reaction of the oxygen with CeO_{2-x} can be written[9] $O_2 \leftrightarrow \frac{1}{2}O_2 + V_O'' + 2e'$. The reaction constant, $K(T)$, is included in the constant in Eq. (3).[9] We find at 1073 K, $K(T) = 3\times10^{-15}$ (compared with 1.5×10^{-15} in Ref. 9). The impurity concentration (assumed to be doubly charged, as for Ca_{Ce}'') can be determined from the constant in Eq. (4) and from $K(T)$. This yields a concentration of ~ 800 ppm in accordance with the direct impurity analysis (900 ppm Ca[10]).

The partial molar enthalpy of oxygen $\Delta\bar{H}(O_2)$ was determined from $\partial[\mu(O_2)/T]/\partial(1/T)$ and found to be 870 kJ/mol (9 eV) for $x \gtrsim 3\times10^{-3}$ comparable to values found by others.[9,12–14] $\Delta\bar{H}(O_2)$ decreases for smaller x to a value of 670 kJ/mol (7 eV) for $x = 10^{-4}$, this was not previously observed.

We have demonstrated that a setup for coulometric titration and EMF measurements for oxygen in a wide $P(O_2)$ and T range can be constructed and accurately and conveniently operated. We used our setup to determine thermodynamic properties of nonstoichiometric ceria (CeO_{2-x}). The measurements confirm previous results,[9,12] that the defects in CeO_{2-x} can be treated using the Boltzmann statistics (for $x \leqslant 10^{-2}$ and $T \geqslant 800$ K). Our data extend the $P(O_2)$ vs x relations in CeO_{2-x} to lower temperatures than measured before.

The authors thank D. S. Tannhauser for helpful remarks.

[1] H. Rickert, *Electrochemistry of Solids* (Springer, Berlin, 1982), pp. 129–148.
[2] H. Schmalzried, Z. Phys. Chem. **38**, 87 (1963).
[3] H. Rickert, *Electrochemistry of Solids* (Springer, Berlin, 1982), p. 120.
[4] J. D. Tretyakov and D. A. Rapp, Trans. Met. Soc. AIME **245**, 1235 (1969).
[5] U. Hölscher and H. Schmalzried, Z. Phys. Chem. **139**, 69 (1984).
[6] C. M. Mari, S. Pizzini, L. Manes, and F. Toci, J. Electrochem. Soc. **124**, 1831 (1977).
[7] Yu. D. Tretyakov, V. F. Komarov, N. A. Prosvirnina, and I. B. Kutsenok, J. Solid State Chem. **5**, 157 (1972).
[8] M. Ricken, J. Nölting, and I. Riess, J. Solid State Chem. **54**, 89 (1984), I. Riess, M. Ricken, and J. Nölting, J. Solid State Chem. **57**, 314 (1985).
[9] H. L. Tuller and A. S. Nowick, J. Electrochem. Soc. **126**, 209 (1979).
[10] Impurity analysis by emission spectroscopy done by KFA, Jülich, indicates that the main impurities are Ca 900 ppm (and various rare-earth atoms at a possible similar total concentration).
[11] P. J. M. Bevan and J. Kordis, J. Inorg. Nucl. Chem. **26**, 1509 (1964).
[12] R. J. Panlener, R. N. Blumenthal, and J. E. Garnier, J. Phys. Chem. Solids **36**, 1213 (1975).
[13] J. W. Dawicke and R. N. Blumenthal, J. Electrochem. Soc. **133**, 904 (1986).
[14] J. Campserveux and P. Gerdanian, J. Chem. Thermodyn. **5**, 795 (1974).

19 March 2026 09:52:58

Buried transverse-junction stripe laser for optoelectronic-integrated circuits

Jun Ohta, Ken'ichi Kuroda, Kazumasa Mitsunaga, Kazuo Kyuma, Kouichi Hamanaka, and Takashi Nakayama
Central Research Laboratory, Mitsubishi Electric Corporation, 8-1-1 Tsukaguchi-Honmachi, Amagasaki, Hyogo 661, Japan

(Received 22 September 1986; accepted for publication 13 January 1987)

A buried transverse-junction stripe (TJS) laser, a suitable laser diode for integration, has been fabricated using molecular-beam epitaxial growth. The fundamental characteristics of the laser were as good as those of a conventional TJS laser. We have also demonstrated a fabrication process for the laser that is compatible with that for metal-semiconductor field-effect transistors (MESFETs) to be monolithically integrated. The characteristics of the MESFETs fabricated on substrates processed for the laser were the same as those fabricated on virgin substrates when such processed substrates were etched 2 μm deep.

Optoelectronic-integrated circuits (OEICs) are promising devices for optical communication and/or signal processing systems. There have been many attempts to integrate both optical and electronic devices monolithically.[1,2] The difficulties in planar integration of these devices mainly arise from the difference in their device geometry and fabrication processes including heteroepitaxy; the thickness of a laser diode (LD) is about 5 μm and that of an electronic device is less than 1 μm. Therefore, the matching of these two kinds of devices is key to the realization of monolithic integration. In order to fully utilize the advantages of a pure planar structure, where the surface of the chip is flat and no steps exist, the laser should be fabricated on a semi-insulating (SI) substrate and also bear both its electrodes on the surface. These requirements can be achieved by introducing a transverse-junction stripe (TJS) laser into the OEICs. Although an integrated laser of the TJS type has already been reported, there has been no clear indication of the availability of this laser in OEICs.[3,4] Moreover, before integrating a TJS laser with electronic devices, such as field-effect transistors (FETs) for driving and modulation, we must examine whether the fabrication process for the laser affects the per-

 0021-8979/87/104933-03$02.40 © 1987 American Institute of Physics